

previously established point.

Illustrated in Fig. 1, we observe a vehicle equipped with lidar technology. Subsequently, Fig. 2 showcases

the ROS environment mapping the surroundings, generating paths (indicated by green lines) on maps through real-time tracking of the

vehicle's movement. This fusion of capabilities empowers the ADAS system to navigate and drive vehicles without collisions by efficiently avoiding obstacles.

To facilitate this advancement, a list of components required for the project is outlined in Table 1. If you have previously constructed the ADAS system as per this series of articles, you likely possess these components.

Before delving into the project, it is essential to familiarise yourself with three key concepts: SLAM, Hector SLAM, and dead reckoning.

SLAM (simultaneous localisation and mapping). It is a fundamental problem in robotics and computer vision, where the goal is for an autonomous system (such as a robot or a self-driving car) to simultaneously build a map of its environment and accurately determine its own position within that map in real time.

Hector SLAM. It is a popular and efficient open source SLAM algorithm designed for mobile robots. The key

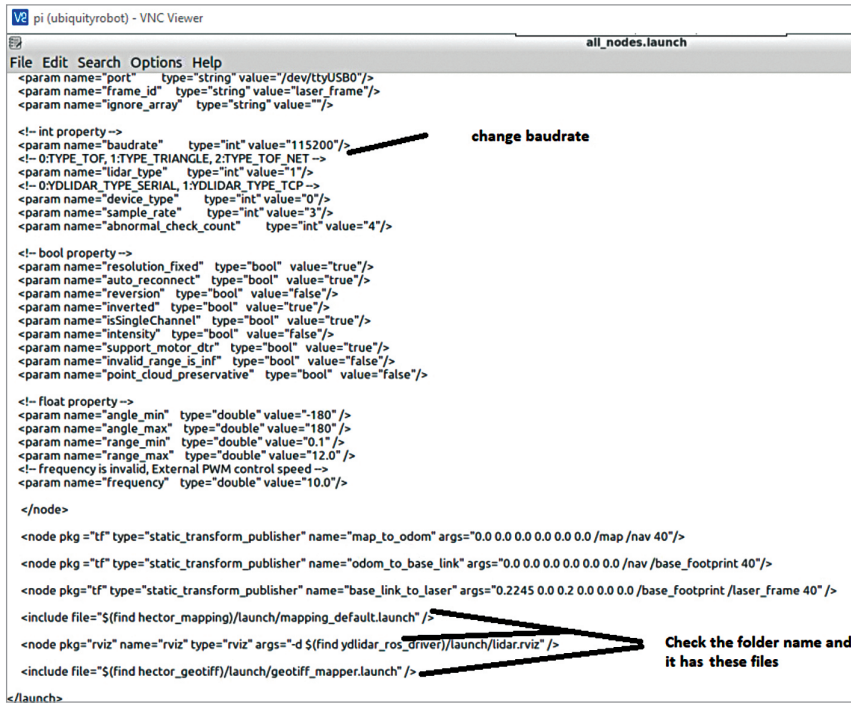


Fig. 3: Configuring the ROS nodes according to lidar

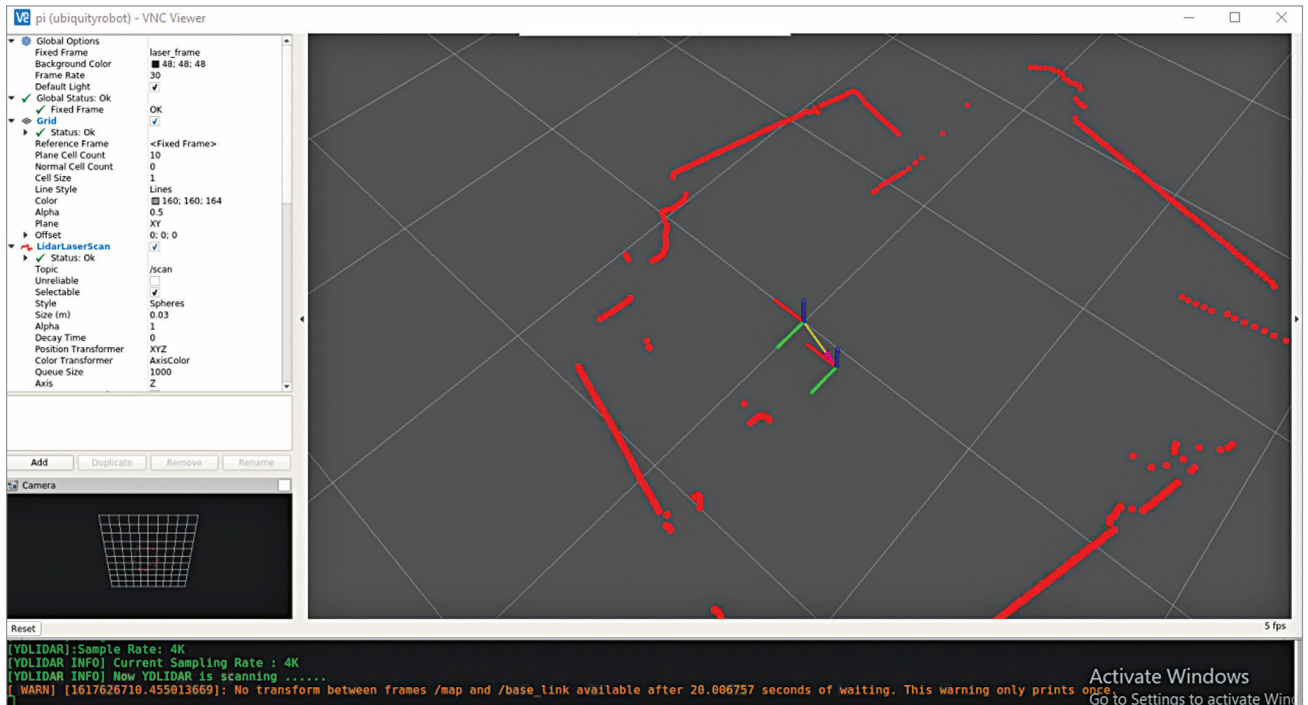


Fig. 4: SLAM data